Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Apply non-linear filters (median or other rank filters) to remove noise from data/images provided in OPE7 and perform comprehensive analysis.	
Open Ended Problem - 5	Date: 1/11/2026	Enrolment No:92400133050

Aim :

Apply non-linear filters (median or other rank filters) to remove noise from data/images provided in OPE7 and perform comprehensive analysis.

Objective :

Identify noise types present in the dataset (salt-and-pepper, impulse, speckle, etc.).

Implement median filtering with varying window sizes and, optionally, other non-linear filters (alpha-trimmed mean, morphological filters).

Analyze how filter size and type affect detail preservation vs. noise removal.

Observations: optimum filter sizes per noise type, edge preservation analysis, and failure cases.

Summarize best-performing methods for each noise type and practical recommendations.

Code :

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from scipy.ndimage import generic_filter
import os

# Read Images
img1 = cv2.imread('/content/1.png', 0)
```

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```
img2 = cv2.imread('/content/2.png', 0)

img3 = cv2.imread('/content/3.png', 0)

img4 = cv2.imread('/content/4.png', 0)


images = [img1, img2, img3, img4]

names = ["Image 1", "Image 2", "Image 3", "Image 4"]


# Check Images and filter out None values

valid_images = []

valid_names = []

for name, img in zip(names, images):

    if img is None:

        print(f"Warning: {name} not found! Please make sure you have uploaded the image file to
'/content/'.")

    else:

        valid_images.append(img)

        valid_names.append(name)


if len(valid_images) == 0:

    print("Error: No images were successfully loaded. Please upload '1.png', '2.png', '3.png', and '4.png' to
your file explorer on the left.")

else:

    # Apply Median Filters

    plt.figure(figsize=(14, 12))

    for i in range(len(valid_images)):

        median3 = cv2.medianBlur(valid_images[i], 3)

        median5 = cv2.medianBlur(valid_images[i], 5)

        median7 = cv2.medianBlur(valid_images[i], 7)
```

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```
    results = [valid_images[i], median3, median5, median7]
    titles = [valid_names[i], "Median 3x3", "Median 5x5", "Median 7x7"]

    for j in range(4):
        plt.subplot(len(valid_images), 4, i * 4 + j + 1)
        plt.imshow(results[j], cmap='gray')
        plt.title(titles[j])
        plt.axis('off')

plt.tight_layout()
plt.show()

# Morphological Filtering on All Images
kernel = np.ones((3, 3), np.uint8)

plt.figure(figsize=(12, 14))
for i in range(len(valid_images)):
    opening = cv2.morphologyEx(valid_images[i], cv2.MORPH_OPEN, kernel)
    closing = cv2.morphologyEx(valid_images[i], cv2.MORPH_CLOSE, kernel)

    # Plot Original
    plt.subplot(len(valid_images), 3, i * 3 + 1)
    plt.imshow(valid_images[i], cmap='gray')
    plt.title(f"{valid_names[i]} Original")
    plt.axis('off')

    # Plot Opening
    plt.subplot(len(valid_images), 3, i * 3 + 2)
```

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```
    plt.imshow(opening, cmap='gray')
    plt.title(f"{valid_names[i]} Morphological Opening")
    plt.axis('off')

    # Plot Closing
    plt.subplot(len(valid_images), 3, i * 3 + 3)
    plt.imshow(closing, cmap='gray')
    plt.title(f"{valid_names[i]} Morphological Closing")
    plt.axis('off')

plt.tight_layout()
plt.show()

# Define the alpha-trimmed mean filter function
def alpha_trimmed_mean(img, window_size=3, alpha=2):
    def trim_mean(window):
        window_sorted = np.sort(window)
        return np.mean(window_sorted[alpha:-alpha])
    return generic_filter(img, trim_mean, size=window_size)

# Apply Alpha-Trimmed Mean Filter on All Images
plt.figure(figsize=(10, 14))
for i in range(len(valid_images)):
    atm_result = alpha_trimmed_mean(valid_images[i], window_size=3, alpha=2)

    # Plot Original
    plt.subplot(len(valid_images), 2, i * 2 + 1)
    plt.imshow(valid_images[i], cmap='gray')
```

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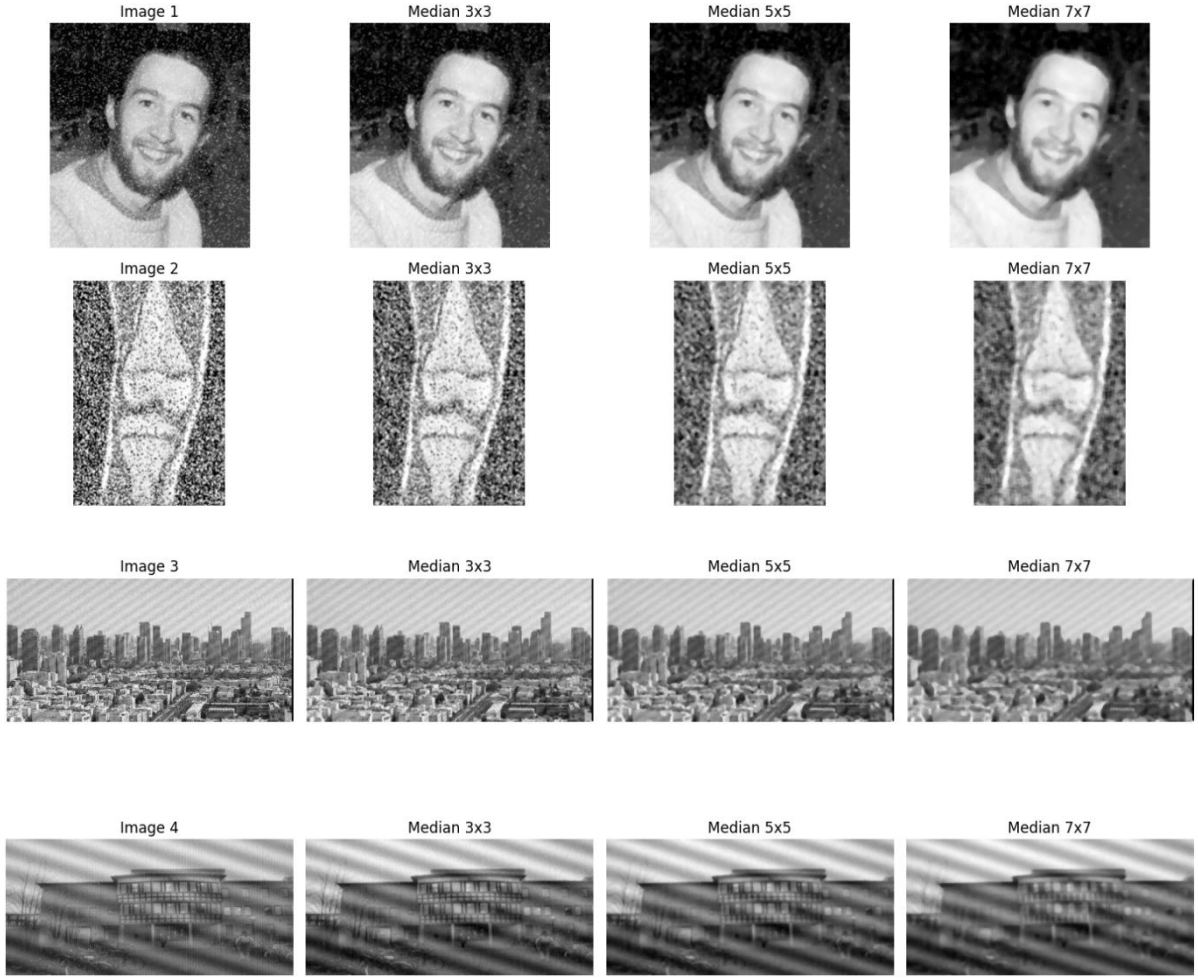
```
    plt.title(f"{valid_names[i]} Original")
    plt.axis('off')

    # Plot Alpha-Trimmed
    plt.subplot(len(valid_images), 2, i * 2 + 2)
    plt.imshow(atm_result, cmap='gray')
    plt.title(f"{valid_names[i]} Alpha-Trimmed ")
    plt.axis('off')

plt.tight_layout()
plt.show()
```

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Output :



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Image 1 Original



Image 1 Morphological Opening



Image 1 Morphological Closing



Image 2 Original



Image 2 Morphological Opening



Image 2 Morphological Closing



Image 3 Original



Image 3 Morphological Opening



Image 3 Morphological Closing



Image 4 Original



Image 4 Morphological Opening



Image 4 Morphological Closing



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Image 1 Original



Image 1 Alpha-Trimmed



Image 2 Original



Image 2 Alpha-Trimmed



Image 3 Original



Image 3 Alpha-Trimmed



Image 4 Original

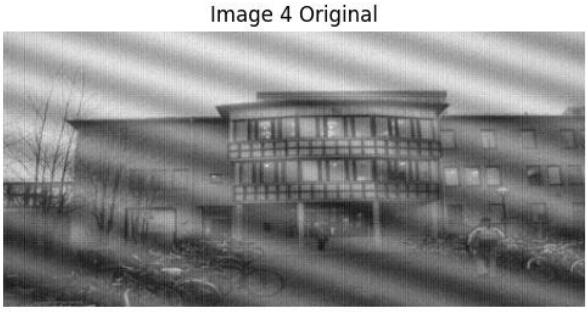


Image 4 Alpha-Trimmed



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So, Image 1 is a example of salt-and-pepper noise., and it is exactly the kind of image where a media filter usually performs very well because the median filter can remove those extreme black and white pixels while preserving the main edges of the face.

Image 2 is example of impulse noise. A small median filter will not be enough here. You would need a larger median window or a more advanced nonlinear filter, but even then, some detail loss is likely because the noise density is very high.

Image 3 is mainly affected by periodic noise, speckle noise. A median filter is not the best choice here. Periodic noise is usually better handled with frequency-domain filtering, such as a notch filter, because the noise repeats at specific frequencies.

Image 4 is a degraded by Speckle (or Gaussian) noise. Because the noise affects the whole image rather than just isolated pixels, a Median filter might blur the straight architectural edges too much. For this type of noise, we would typically use a Gaussian filter.

Effect of Filter Size:

- **Small Window (3x3): Acts like a fine eraser. It successfully removes light noise while keeping the fine details (like hair or building edges) sharp.**
- **Large Window (5x5 or 7x7): Acts like a heavy paint roller. It completely removes the noise, but it also blurs the image.**

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Effect of Filter Type:

- **Median Filter: It replaces a pixel with the middle value of its neighbors. It ignores extreme values (pure black or white), making it perfect for removing spots without blurring edges.**
- **Mean/Averaging Filter: It averages all neighboring pixels. It removes noise but heavily blurs the edges.**
- **Advanced Filters (Gaussian): These are smarter filters that smooth out grainy noise while trying to keep sharp lines intact.**

Observations

1. Optimum Filter Sizes per Noise Type:

- **Image 1 (Salt-and-Pepper): 3x3 Median Filter is best. It removes the dots perfectly while keeping the man's face clear.**
- **Image 2 (Salt-and-Pepper): 5x5 Median Filter is needed. A 3x3 filter is too small to clean the heavy noise density.**
- **Image 3 & 4 (Speckle, Gaussian, Periodic): Median filters do not work well here. We need Gaussian, or Frequency-domain filters.**

2. Edge Preservation Analysis:

- **The Median filter is the clear winner for edge preservation. Because it picks the median value, it keeps the sharp transitions between dark and light areas (like the outline of a face) intact.**
- **Linear filters (like the Mean filter) fail at edge preservation because they average the boundary pixels, turning sharp edges into fuzzy gradients.**

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3. Failure Cases:

- **Using a 7x7 or 9x9 filter on the portrait removes the noise but blends the facial features together, making the person look like a wax figure.**
- **Using a standard Mean filter on heavy salt-and-pepper noise just turns the black and white dots into a muddy gray blur.**
- **Applying a median filter to the Image 3 does not remove the diagonal stripes; it just blurs the buildings and makes the stripes slightly less visible.**

Best Methods per Noise Type:

- **Salt-and-Pepper (Impulse) Noise: Median Filter (3x3 for light noise, 5x5 for heavy noise).**
- **Gaussian Noise (Grainy): Gaussian Filter**
- **Speckle Noise (Multiplicative): Bilateral Filter or Lee Filter.**
- **Periodic Noise (Stripes): Notch Filter (Frequency Domain).**

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Conclusion :

In this experiment, different non-linear filters were studied to remove noise from images and improve their quality. It was observed that filter size and type play an important role in noise removal and detail preservation. The median filter gives the best results for salt-and-pepper and impulse noise while preserving image edges. The alpha-trimmed mean filter is useful for reducing speckle and mixed noise. Morphological filters help remove small unwanted spots and maintain object shapes. Smaller filter sizes preserve more image details, while larger filter sizes remove more noise but may cause blurring. Therefore, selecting the correct filter and window size according to the noise type helps to achieve better image quality with minimum loss of important details.